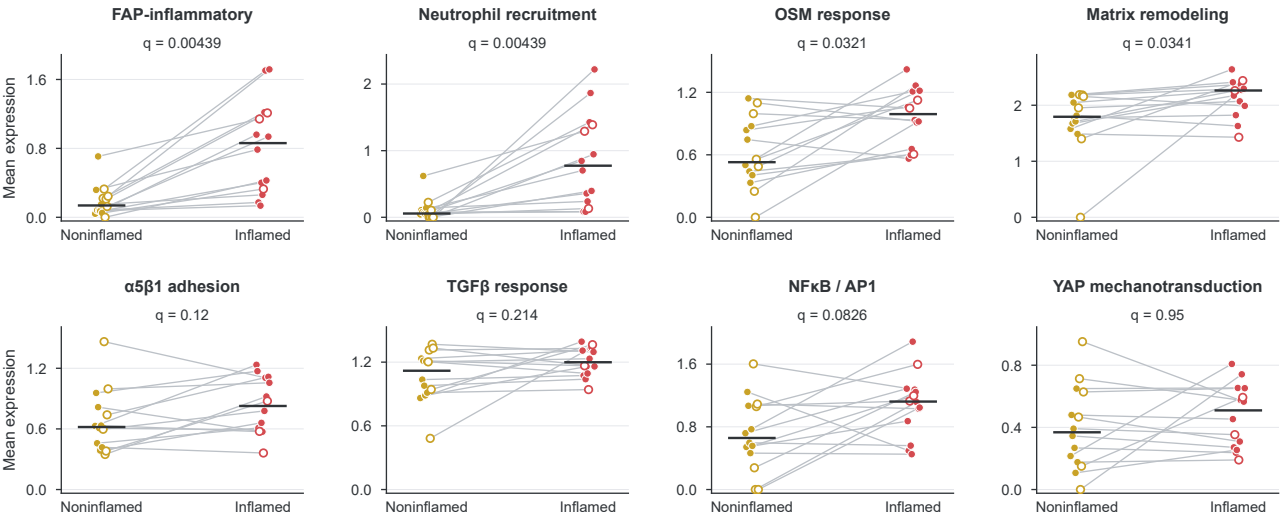


Supplementary Figure 1

A Paired stromal expression of fibroblast-associated programs

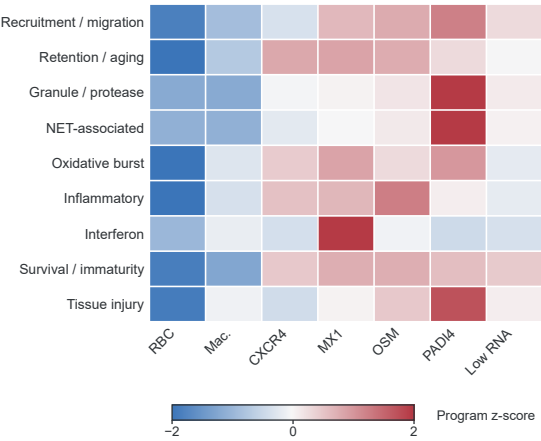
Matched noninflamed and inflamed UC tissue; 14 patients



Paired Wilcoxon q (BH, 8 programs). Lines: patients; bars: medians; open points: <10 recovered cells.

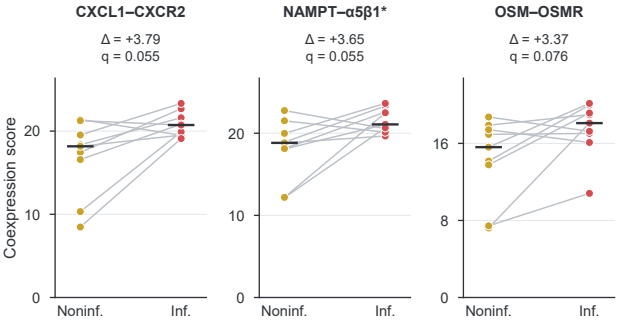
B Pooled RNA programs and annotation controls

All conditions; original gene modules



C Paired ligand–receptor coexpression

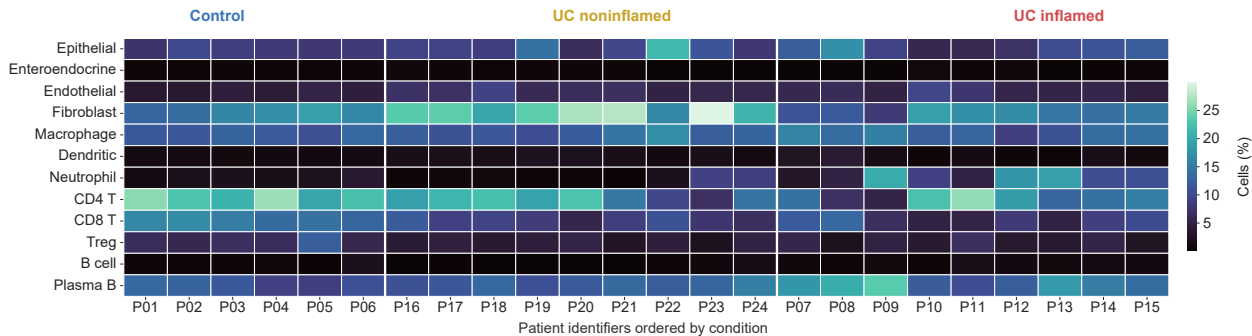
9 matched UC patient pairs



Paired Wilcoxon q (BH, 5 original routes).

*Receptor summary: min(ITGA5, ITGB1).

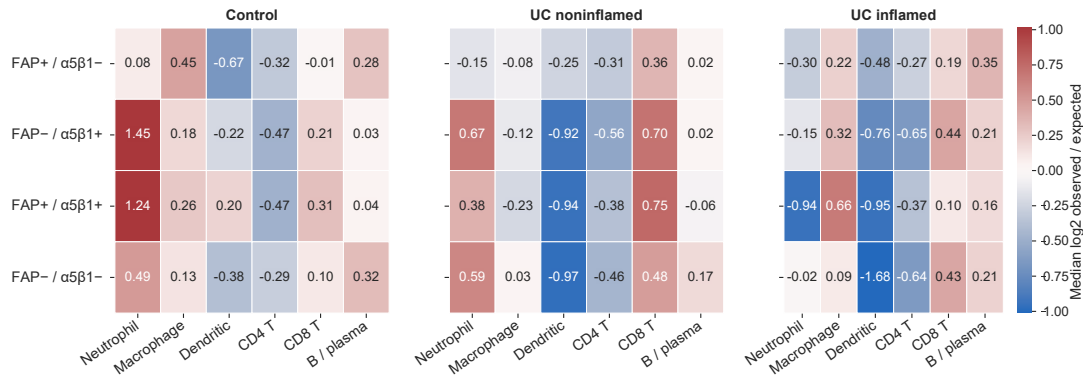
Δ : median paired change; lines: patients.



Supplementary Figure 3

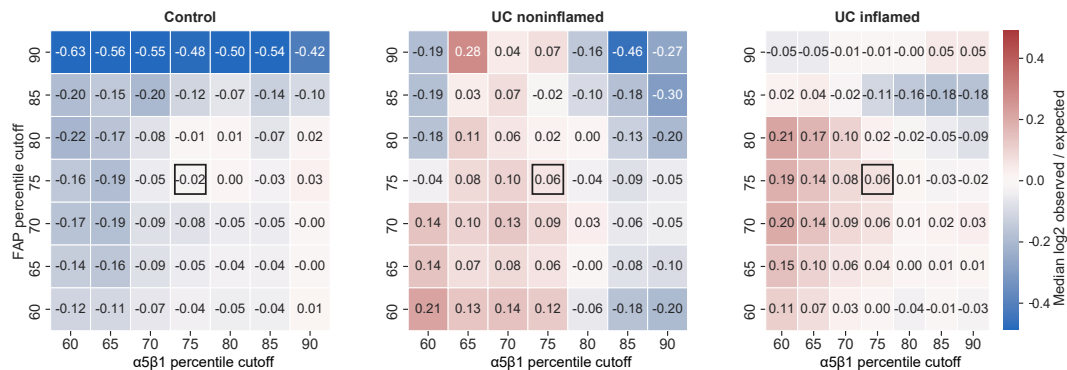
A Immune-lineage spatial specificity

Median log2 observed/expected proximity across FAP/ $\alpha 5\beta 1$ strata



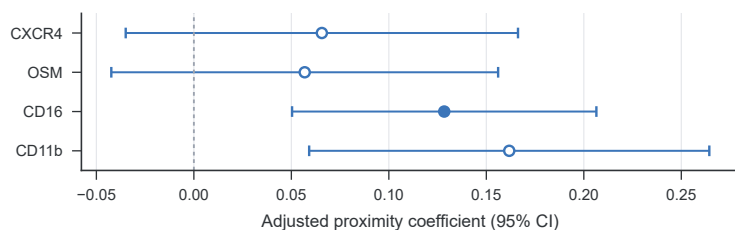
B Threshold sensitivity

FAP and $\alpha 5\beta 1$ percentile cutoffs across tissue conditions



C Adjusted neutrophil proximity effects

Filled point: restricted spatial-null $q < 0.05$; bars show 95% CI



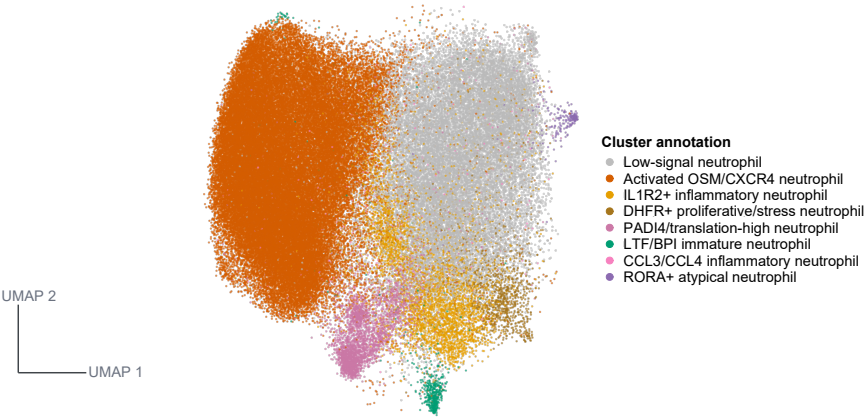
Model q	Spatial-null q
0.242	0.844
0.242	0.843
0.008	0.032
0.008	0.144

Figure 2 is a dot plot showing the average expression and percent expressed of 20 genes across four cell types. The y-axis lists the cell types: S100A8/A9+ immune-like, CCN1/CCN2+ activated, CSF3R+ immune-like, and COL1A1+ inflammatory ECM. The x-axis lists the genes: COL1A1, COL1A2, COL3A1, DCN, LUM, FAP, IL6, CXCL8, CCN1, CCN2, SPARC, COL2, FOXF1, TFP1, ACTA2, TAGLN, CSF3R, C5AR1, TYROBP, S100A8, S100A9, CLEC12A, and SIGLEC14. The color of the dots represents the average expression, ranging from -1.0 (blue) to 1.5 (red). The size of the dots represents the percent expressed, with sizes corresponding to 0, 25, 50, 75, and 100%.

Supplementary Figure 5

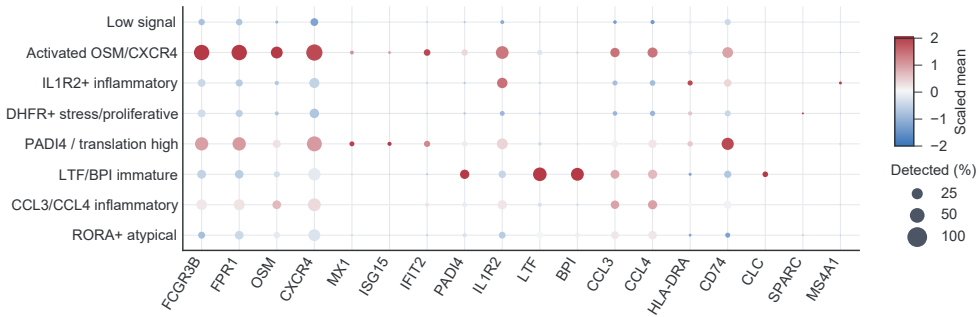
A Neutrophil subcluster atlas

Retained cells colored by transcriptomic subcluster annotation



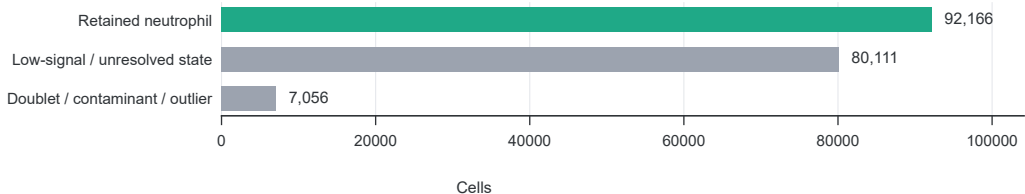
B Marker support for subcluster annotations

Dot size: detected cells; color: scaled mean expression



C Cell-retention and cleanup inventory

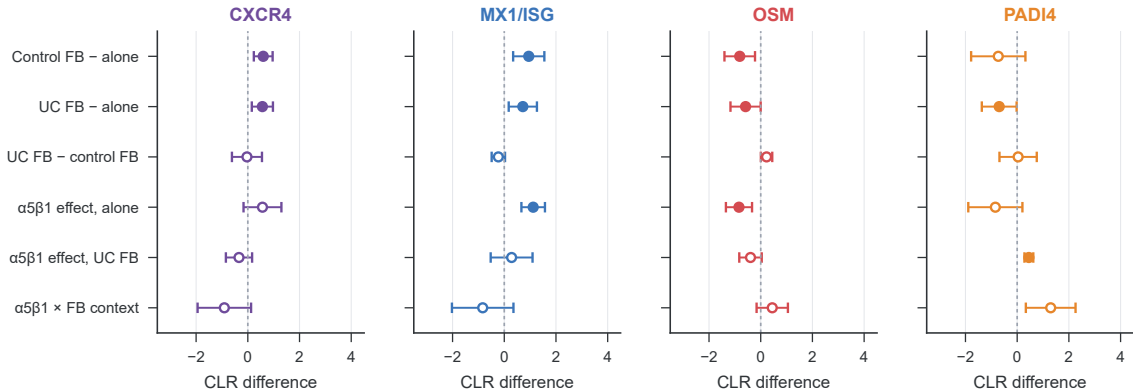
Original counts summarized by disposition



Supplementary Figure 6

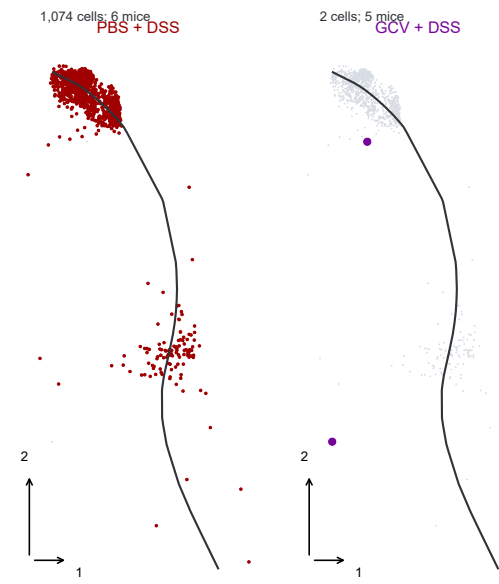
A Recorded-donor neutrophil state contrasts

Six recorded donors; filled points denote BH $q < 0.05$; bars show 95% CI

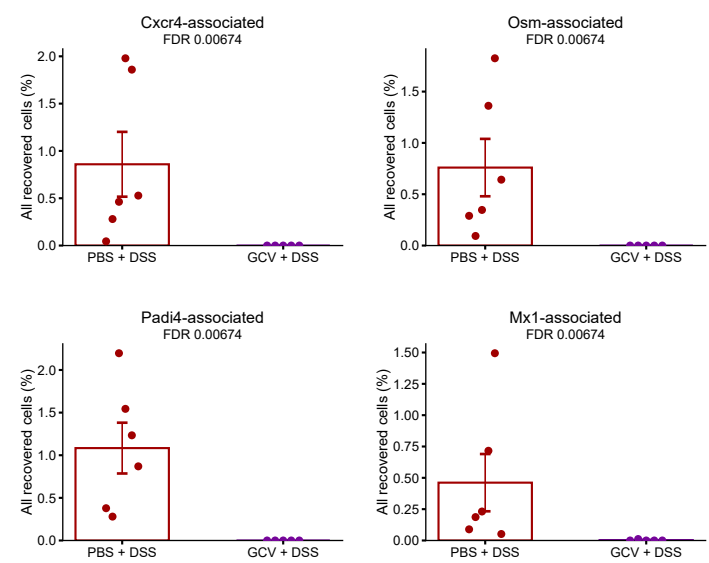


Supplementary Figure 7

A Neutrophil occupancy after FAP ablation

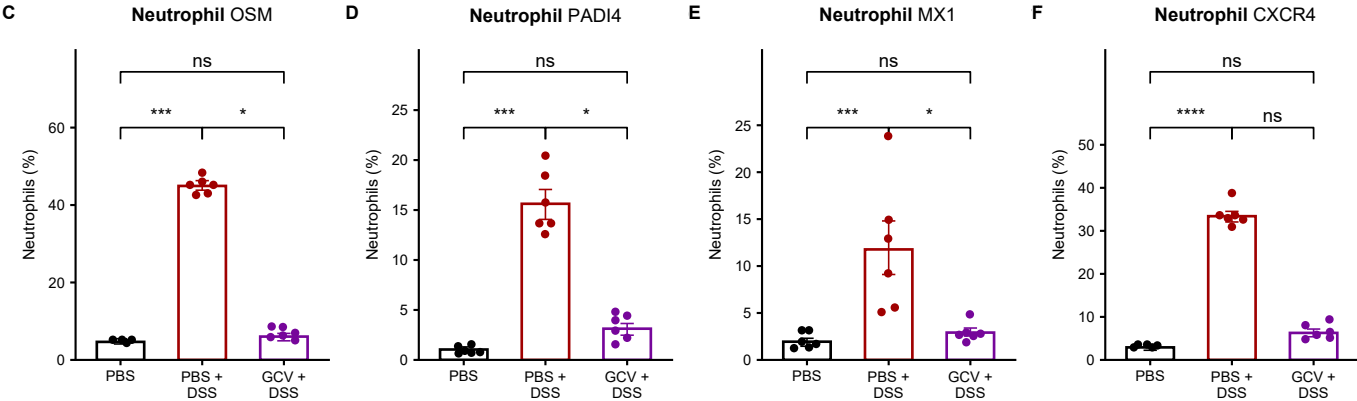


B Recovered neutrophil-state fractions



Only two postablation neutrophils: occupancy loss, not a postablation trajectory or within-state estimate.

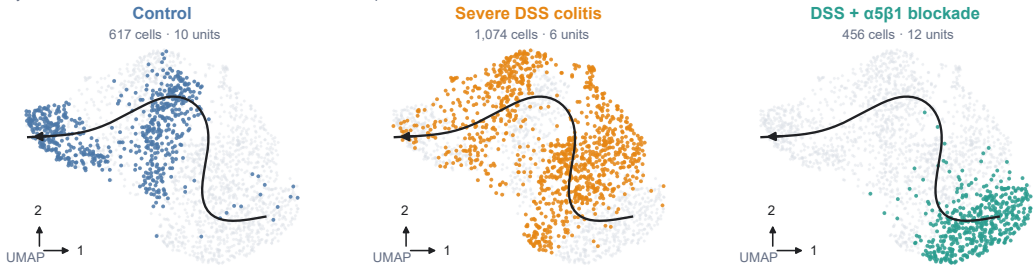
C-F Neutrophil marker readouts supporting Figure 4



Supplementary Figure 8

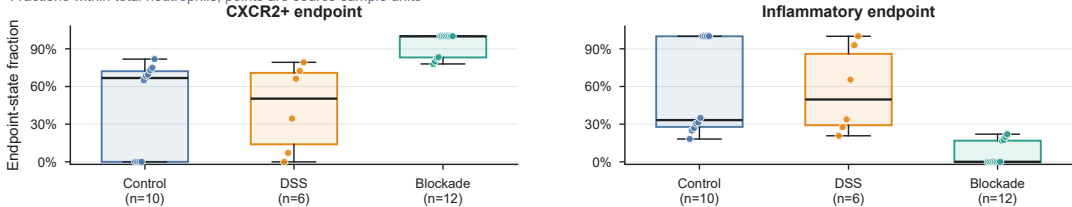
A $\alpha 5 \beta 1$ blockade redistributes occupancy along a shared neutrophil continuum

Grey denotes the full reference manifold; arrow indicates imposed orientation, not measured time



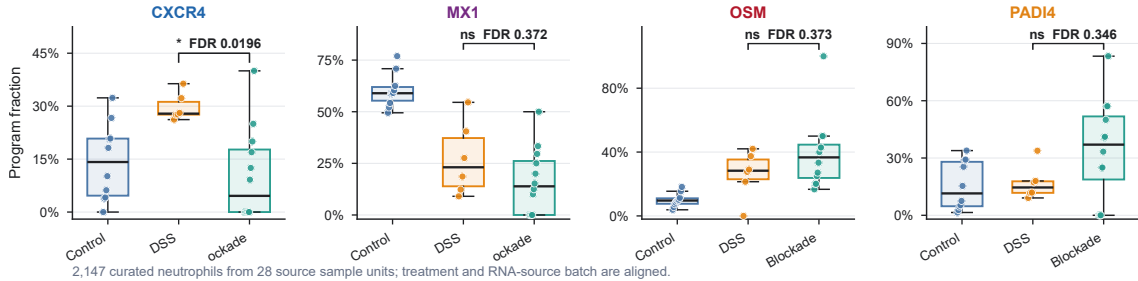
B Endpoint-state redistribution

Fractions within total neutrophils; points are source sample units



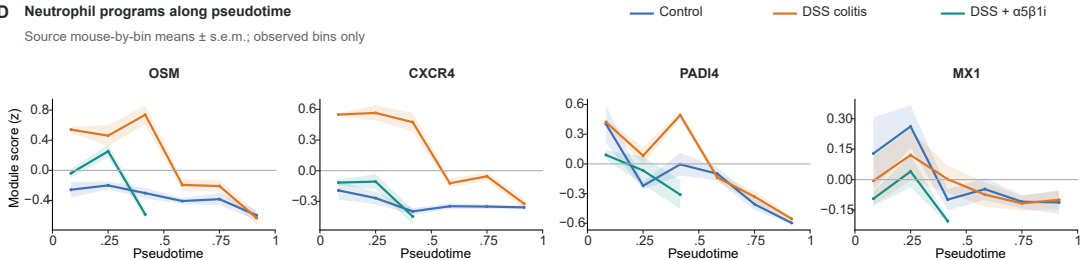
C Transferred human-associated neutrophil programs

Ortholog-mapped dominant program; source DSS-versus-blockade tests retained



D Neutrophil programs along pseudotime

Source mouse-by-bin means $\pm$ s.e.m.; observed bins only



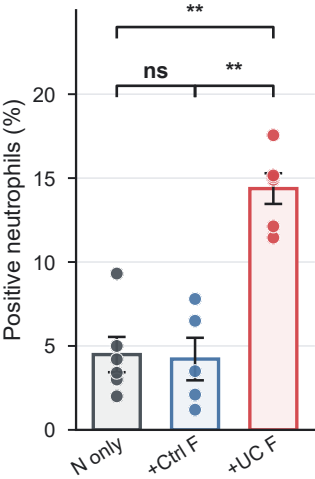
Supplementary Figure 9

Human fibroblast coculture: neutrophil protein readouts

Neutrophils only + control fibroblasts + UC fibroblasts

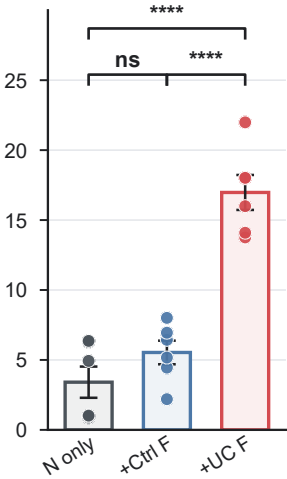
A

PADI4+ neutrophils



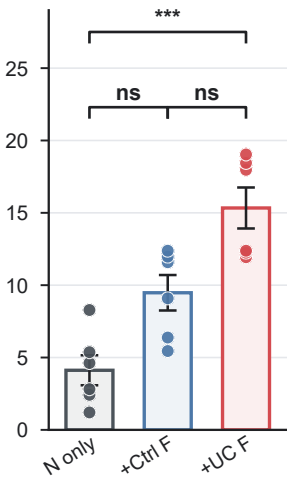
B

OSM+ neutrophils



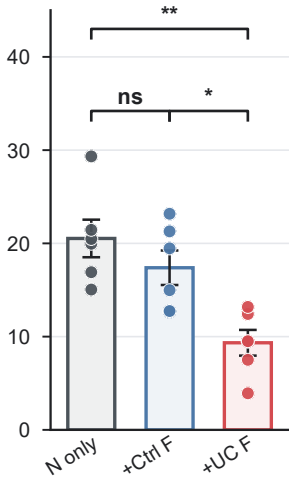
C

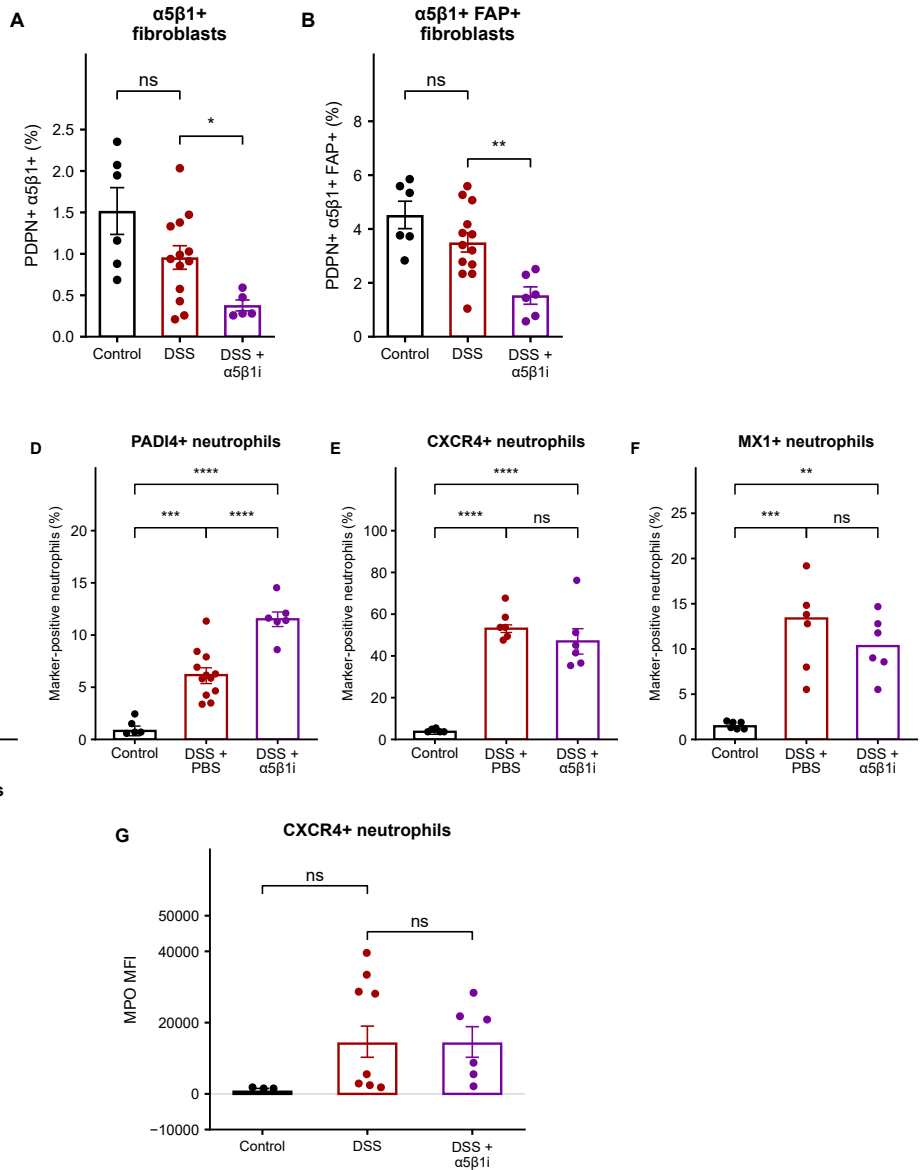
MX1+ neutrophils



D

CXCR4+ neutrophils

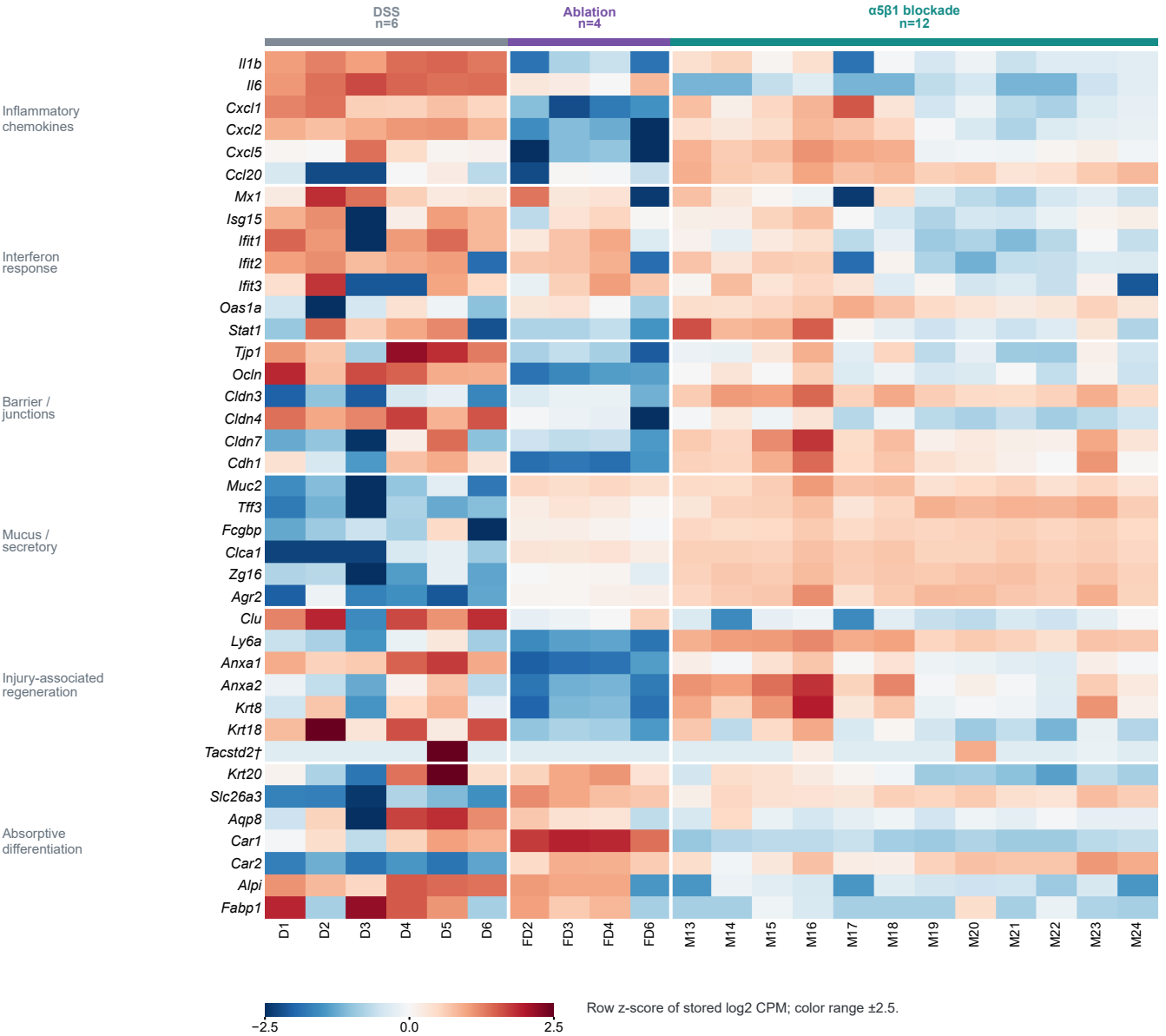




Supplementary Figure 11

Complete epithelial program gene expression

Mature absorptive colonocytes; each column is one biological mouse



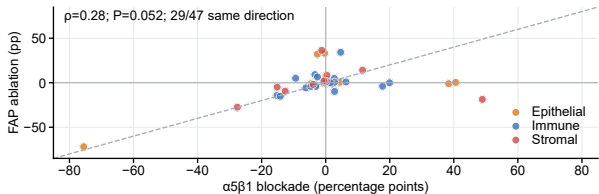
†Tacstd2 detected in 4/22 mice. Descriptive display; no gene-level hypothesis tests.

Supplementary Figure 12

Cell-state concordance and predicted communication

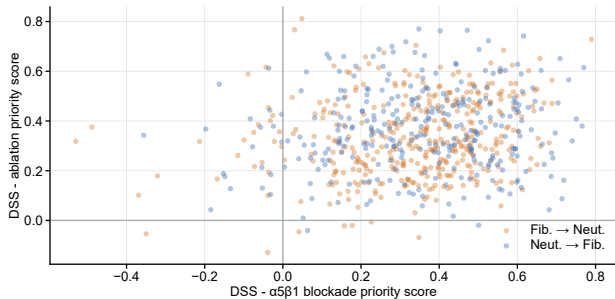
Separate stromal interventions compared with a shared DSS reference

A Cell-state response concordance



B Predicted interaction concordance

691 routes: $\rho=0.15$, $P=0.00013$; core: 490/502 attenuated



Panel C: intervention key

● FAP-lineage ablation

▲ $\alpha 5 \beta 1$ blockade

Positive values indicate lower predicted communication priorities relative to DSS.

C Predicted reciprocal routes

